

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0290

Roy C Miller 138 kV
July 25, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0290. The Interconnection Customer proposes a 228.1 MW large load (data center) facility interconnecting at the Roy C Miller 138 kV substation in the New Jersey area. The request was received on November 19, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,001,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of Roy C Miller 345/138 kV Transformer Bank, the Eta - Kilner 138 KV loads to 152.8% of its 143 MVA rating. Project DFAX: 61.9%.

For the loss of Roy C Miller 345-138 kV Transformer Bank, the Watertown - S Mercer 138 KV loads to 109.6% of its 243 MVA rating. Project DFAX: 100.0%.

For the loss of Merret - Sunrise Ctr Substation 500 KV, the Wahiawa - N Allegheny Windpower Project 345 KV loads to 102.8% of its 1,224 MVA rating. Project DFAX: 5.5%.

The Ironwood Llc - Fayette 345 KV is overloaded to 100.8% of its 1,244 MVA emergency rating for the loss of the Upsco - Wigwam 345 KV line. The project contributes 5.1% DFAX (11.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Eta - Kilner 138 KV	Rebuild line with higher-capacity conductor	\$3,946,000	100% (\$3,946,000)
Watertown - S Mercer 138 KV	Reconductor line	\$415,000	100% (\$415,000)
Wahiawa - N Allegheny Windpower Project 345 KV	Reconductor line	\$10,023,000	4% (\$395,000)
Ironwood Llc - Fayette 345 KV	Reconductor line	\$7,579,000	3% (\$245,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$5,001,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.